# Sets, Limits of sequences of sets and monotone classes

Let $\Omega$ be a non empty set and let $A_1, A_2, A_3, \ldots$ be a Sequence of subsets of $\Omega$.

**Def: [Increasing sequence of sets]**

If $A_1 \subseteq A_2 \subseteq A_3 \cdots$, then we say that the sequence $\{A_n\}_n$ is non-decreasing or increasing. In this case, we write $A_n \uparrow$. If $A = \bigcup_{n=1}^{\infty} A_n$ we say $\{A_n\}_n$ is increasing to $A$ and denote it has $A_n \uparrow A$.

**Definition: [Decreasing sequence of sets].**

If $A_1 \supseteq A_2 \supseteq A_3 \cdots$, then we say that the sequence $\{A_n\}_n$ is non-increasing or decreasing. In this case, we write $A_n \downarrow$. If $\bigcap_{n=1}^{\infty} A_n = A$, then we say $\{A_n\}_n$ is decreasing to $A$ and denote it $A_n \downarrow A$

* If $A_n \uparrow A$ iff $A_n^c \downarrow A^c$

Example: i) for a non-empty set $\Omega = \mathbb{R}$, take $A_n = (-n, n)$ $n = 1, 2, \ldots$ then $A_n \uparrow \mathbb{R}$.

(ii) for $\Omega = \mathbb{R}$ fix $x \in \mathbb{R}$ and consides $A_n = (-\alpha, n - \frac{1}{n})$, $n = 1, 2 \ldots$

then $A_n = (-\infty, x - 1/n]$ $n = 1, 2, \ldots$

then $A_n \uparrow (-\alpha, x)$.

(iii) for $\Omega = \mathbb{R}$ fix $x \in \mathbb{R}$ and consides

$A_n = (-\infty, x + 1/n]$, $n = 1, 2, \ldots$

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then $A_n \downarrow (-\infty, x]$.

Monotone class :

A collection of $\mathcal{F}$ of subsets of $\Omega$ is called a monotone class if it is closed under countable increasing union and countable decreasing intersection.

i.e., i) If $\{A_n\}_n$ is an increasing sequence of sets in $\mathcal{F}$ and $A_n \uparrow A$, then $A \in \mathcal{F}$

ii) If $\{A_n\}_n$ is a decreasing sequence of sets in $\mathcal{F}$ and $A_n \downarrow A$. then $A \in \mathcal{F}$.

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In the introduction, we recall notion of Set of all outcomes of a random experiment, i.e., Sample space We also observed that the events are special subsets of the sample space.
Moreover, the collection of all events satisfy some algebraic properties. In this lecture, we study these properties and model the collection of all events as "$\sigma$-field". In this regard, note that the sample space, by construction, is a non-empty set.

To discuss the notion of $\sigma$-fields, $\sigma$ algebra We shall work with a non-empty set $\Omega$. In reference to a random experment, the Set $\Omega$ is a taken as the sample space.

Classes of Sets : A group of sets will be turned as Classes of Sets or class.

Rings of Sets : A Ring $R$ is a non-empty class of sets which is closed under the formation of "finite unions" and difference.

i.e.,
If $A \in R$, $B \in R$ then $A \cup B \in R$ and $(A - B) \in R$.

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§ Ex: $R = \{A \subseteq N : A \text{ is finite}\}$ i.e., finite subset of Natural number $|N|$ is a Ring.

Proof: First we have prove $A \cup B \in R$ first condition of ring as finite union of finite set is again finite.

$\Rightarrow A \in R$ and $B \in R$ (finite set) of R then

$(A \cup B) \in R$.

As difference of subset of finite set is finite

$\Rightarrow (A - B) \in R$.

Hence R is a ring of sets.

§ Ex. Power set $P(x) = \{\phi, X\}$ of a set X is a ring of sets.

Proof: Since $A \cup B \in P(x)$

also $A - B \in P(x)$.

Let $A = \phi$ $B = \phi$

$A = \phi$ $B = X$

$A = X$ $B = \phi$

$A = X$ $B = X$.

$A \cup B = \phi$, $A \cup B = X$

$A \cup B = X$ $A \cup B = X$

$\therefore$ $P(x)$ is a ring of sets.

* Ex: Null set $\phi$ is a ring of sets

Sime $\phi \cup \phi = \phi \in R$.

$\phi - \phi = \phi \in R$.

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Sigma Ring: A $\sigma$-ring $C$ is a non-empty set $\sigma$-Ring (class of sets) which is closed under the formation of "Countable Union" and difference"

thus

(i) $A_i \in C,\ i=1,2,\ldots \Rightarrow \bigcup_{i=1}^{\infty} A_i \in C$

(ii) $A \in C,\ B \in C \Rightarrow (A-B) \in C$ or $A \cap B^c \in C$.

Example: let R be the collection of all finite subsets of N. Thus R is a ring but not a $\sigma$-ring.

Algebra of sets: A collection of subsets of a Set X is called an algebra of Sets or Boolian algebra if for any $A, B \in \mathcal{A}$ hence $(A \cup B) \in \mathcal{A}$ and $A^c \in \mathcal{A}$ when A is in $\mathcal{A}$ $x \in \mathcal{A}$

Ex: let $X = \{1, 2\}$ define $\mathcal{A} = \{\phi, X\}$ then $\mathcal{A}$ is an algebra of sets or field or field.

1. $x \in \mathcal{A}$
2. if for an $\phi \in \mathcal{A}$ then $\phi \cup X = x \in \mathcal{A}$ or $x \cup \phi = x \in \mathcal{A}$
2. $\phi^c = x \in \mathcal{A}$ $X^c = \phi \in \mathcal{A}$.

$\mathcal{A}$ Satisfy all three conditions of algebra of set

$\mathcal{A}$ is an algebra of sets.

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Ex: 1. Let $X = \{1, 2\}$ define $\mathcal{A} = \{\phi, X\}$ then $\mathcal{A}$ is an algebra of sets or field.

Verification

(1) $x \in \mathcal{A}$

(2) if for any $\phi \in \mathcal{A}$, $x \in \mathcal{A}$ then $\phi \cup X = x \in \mathcal{A}$
or $x \cup \phi = x \in \mathcal{A}$

(3) $\phi^c = x \in \mathcal{A}$ $x^c = \phi \in \mathcal{A}$

$\mathcal{A}$ satisfies all three conditions of algebra of sets therefore $\mathcal{A}$ is an algebra of sets.

Ex: $X = \{a, b\}$ then power set $P(x) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$. is an algebra of sets

Verifications

1. $x \in P(x)$ or $X \in P(X)$.
2. if for any subset of $X$, $P(X)$ contains all subsets of $X$ and it is closed under finite union and complements

i.e., $\phi \cup \{a\} \cup \{b\} \cup \{a, b\} = \{a, b\} \in P(x)$

and $\phi^c = \{a, b\} \in P(x)$

$\{a\}^c = \{b\} \in P(x)$

$\{b\}^c = \{a\} \in P(x)$

$\{a, b\}^c = \phi \in P(x)$

$P(x)$ Satisfies all three conditions of algebra of sets

hence, $P(x)$ is an algebra of sets.

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# σ Algebra or σ field:

If we toss a coin twice the out comes

$\{HH, HT, TH, TT\}$.

Let X: total no of Heads.

$$X = \begin{cases} 0 & \text{Comes up } TT \\ 1 & \text{Comes } HT, TH \\ 2 & \text{comes } HH \end{cases}$$

So Sample space is $S=\{0,1,2\}$ of X. $S^c = \phi$.

Sigma field is nothing but Sample space + collection of Subsets of ~~collect~~ Sample Space

collection of subsets of sample space (power set)

let B is the collection of subsets ie power set.

$$B = \{\{0,1,2\}, \{0\}, \{1\}, \{2\}, \{0,1\}, \{1,2\}, \{0,2\}, \phi\}.$$

We know that

1. A is a non empty set $A \in \sigma$ field $\Rightarrow A^c \in \sigma$ field.
2. If An are countably infinetly many subsets from the $\sigma$ field, then the inter section and Unions of all these is $\sigma$-field.

in B ~~any~~ ~~the~~ all elements are satisfying.

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Let $X$ be a non empty set.

A collection $\mathcal{F} \subset 2^X$ is said to be a $\sigma$-field if it satisfies the following conditions

$2^{\wedge}$ = Powerset

1. $X \in \mathcal{F}$
2. $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$ where $A$ is the subset of $\mathcal{F}$
3. $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

   i.e, closed under countable union. $\left[\left[\bigcup_{i=1}^{\alpha} A_i\right]^{c_\alpha} = \bigcap_{i=1}^{\alpha} A_i\right]$

Ex: Let $X = \{a, b, c\}$ define $\mathcal{F} = \{\phi, X\}$ then s.t $\mathcal{F}$ is $\sigma$-field on $X$.

Sol: Since $x \in \mathcal{F}$ and $\phi^c = X \in \mathcal{F}$.

$X^c = \phi \in \mathcal{F}$

The union of any countable sub collection is either $\phi$ or $X \Rightarrow \mathcal{F} = \{\phi, X\}$ satisfies all three conditions of $\sigma$-algebra, hence $\mathcal{F}$ is an $\sigma$-field.

The term $\sigma$-algebra is equallent to $\sigma$-field.

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Proposition: Let $\mathcal{F}$ be a $\sigma$-field of subsets of $\Omega$ ($\Omega$ sample space) then the following statements are true.

(i) $\Omega \in \mathcal{F}$

(ii) closed under finite union

If $A_1, A_2 \ldots A_n \in \mathcal{F}$ then $\bigcup_{i=1}^{n} A_i \in \mathcal{F}$

(iii) closed under finite Intersection.

If $A_1, A_2 \ldots A_n \in \mathcal{F}$ then $\bigcap_{i=1}^{n} A_i \in \mathcal{F}$.

Ex: Let $X = \{a, b, c\}$ define $\mathcal{F} = \{\phi, X\}$ then s.t. $\mathcal{F}$ is a $\sigma$-field on $X$.

Since $\mathcal{F}$ and $\phi^c = X \in \mathcal{F}$

$X^c = \phi \in \mathcal{F}$.

The Union of any countable sub collections is either $\phi$ or $X \Rightarrow \mathcal{F} = \{\phi, X\}$ satisfy all three conditions satisfied.

Ex: Let $X = \{1, 2, 3\}$ define $\mathcal{F} = \{\phi, \{1\}, \{2,3\}, X\}$.

Show that $\mathcal{F}$ is an $\sigma$-algebra.

Proof: (i) $X \in \mathcal{F}$.

(ii) $\phi^c = X$, $X^c = \phi$, $\{1\}^c = \{2,3\}$, and

$\{2,3\}^c = \{1\} \in \mathcal{F}$.

all complemented elements belongs to $\mathcal{F}$

i.e, $\mathcal{F}$ is closed under Complement.

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3. $\mathcal{F}$ is closed under Countable union
i.e., $\phi \cup \{1\} \cup \{2,3\} \cup X = X \in \mathcal{F}$.

$\therefore$ $\mathcal{F}$ is a satisfied all three conditions,
Hence, $\mathcal{F}$ is an $\sigma$-algebra.

Ex: 3. let $X = N$ Set of natural Numbers.
and $\mathcal{F}$ be the collection of all subsets of $N$ that are either finite of cofinite then $\mathcal{F}$ is an $\sigma$-algebra.

Proof:
(i) $N \in \mathcal{F}$, $\phi \in \mathcal{F}$
(ii) Complement of finite is co-finite and complement of co-finite is finite.
$\Rightarrow$ $\mathcal{F}$ is closed under Complement
(iii) Countable union of finite sets is countable.

$\therefore$ $\mathcal{F}$ is an $\sigma$-algebra / $\sigma$-field.

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**Theorem.** Let $\mathcal{F}_1$ and $\mathcal{F}_2$ are two $\sigma$-algebras on a set $X$ then $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$ also a $\sigma$-field on $X$.

**Proof:** To prove $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$ is $\sigma$-field for this we have to shows that $\mathcal{F}_1 = \mathcal{F}_1 \cap \mathcal{F}_2$ satisfies 3 properties of $\sigma$-field.

**Property 1.** We have to shows that $X \in \mathcal{F}$.

Since $X \in \mathcal{F}_1$ and $X \in \mathcal{F}_2$

$X \in \mathcal{F}_1 \cap \mathcal{F}_2$ (as $X \in \mathcal{F}_1$ and $X \in \mathcal{F}_2$ then $\mathcal{F}_1 \cap \mathcal{F}_2 \in X$).

**Property 2.** We have to $A \in \mathcal{F}_1$ then $A^c \in \mathcal{F}$.

Let $A \in \mathcal{F} \Rightarrow A \in \mathcal{F}_1$ and $A \in \mathcal{F}_2$

Since $\sigma$-algebra are closed under complements

$A^c \in \mathcal{F}_1$ and $A^c \in \mathcal{F}_2$

$A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$.

$\mathcal{F}$ is closed under Complement.

**Property 3.** Closed under the countable union.

Let $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{F} \Rightarrow A_n \in \mathcal{F}_1$ and $A \in \mathcal{F}_2$ $\forall n$

Since $\mathcal{F}_1$ and $\mathcal{F}_2$ are $\sigma$-algebras

$\bigcup_{n=1}^{\infty} \mathcal{F}_1$ and $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_2$

$\Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_1 \cap \mathcal{F}_2 = \mathcal{F}$.

$\mathcal{F}$ is closed under countable unions.

Hence $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$ is an $\sigma$-field.

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# Relation between ring, $\sigma$-ring, field, $\sigma$-field.

1. Every $\sigma$-ring is a ring but Converse is not true in general.

Reason: A $\sigma$-ring is by definition, closed under finite union and set difference, just like a ring.

but Converse is not true, Counter Example

Let R be the collection of all finite subsets one N.

R is closed under finite union and set difference

$\Rightarrow$ R is a ring.

But not closed under countable union

i.e, $A_n = \{n\} \in R$, but $\bigcup_{n=1}^{\infty} A_n = \mathbb{N} \notin R$.

(ii) every $\sigma$-algebra is an algebra but not a $\sigma$-algebra

Counter example for algebra is not $\sigma$-algebra.

Let $X = \mathbb{N}$ (The set of natural numbers).

define the collection

$\mathcal{A} = \{A \subseteq \mathbb{N} \mid A \text{ is finite or } A^c \text{ is finite}\}$ is a algebra

but not a $\sigma$-algebra.

Proof $\mathcal{A}$ is an algebra.

(i) Contains $\mathbb{N}$ and $\emptyset$, $\emptyset$ is finite $\Rightarrow \in \mathcal{A}$

$\mathbb{N}^c = \emptyset \Rightarrow$ co-finite

$\Rightarrow \in \mathcal{A}$

(ii) closed under complement.

if A is finite, then $A^c$ is co-finite

if A is co-finite, then $A^c$ is finite.

$\Rightarrow A^c \in \mathcal{A}$

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closed under finite unions.
Since union of two finite sets is finite
union of a finite and co-finite set is cofinite,
union of two co finite sets is cofinite

$\Rightarrow A \cup B \in \mathcal{A}$

but $\mathcal{A}$ is not a $\sigma$-algebra (field) because
$\mathcal{A}$ is not a closed under countable union.

(iii). Every $\sigma$-algebra is also a ring but converse is not true.

Let $\mathcal{F}$ be a $\sigma$-algebra on a set $X$, we want to show it satisfies properties of $\sigma$-ring.

(1) first property for a $\sigma$-ring because $\mathcal{F}$ is closed under countable union by defn. of $\sigma$-algebra.

(2) $\mathcal{F}$ is closed under set difference.
i.e, $A, B \in \mathcal{F}$, since $\mathcal{F}$ is closed complements and intersections

$$A - B = A \cap B^c \mathcal{F} \Rightarrow \mathcal{F} \text{ is a ring}$$

But $\sigma$-ring not $\sigma$-field.
Counter example: Let $X = \mathbb{R}$, and let $\mathbb{R}$ be the collection of all lebesque measurable. Sets of finite measure.

Sine $\mathbb{R}$ is $\sigma$-ring But $X = \mathbb{R}$ has infinite measure

$\Rightarrow n \notin \mathbb{R} \Rightarrow \mathbb{R}$ is not a $\sigma$-algebra

Note: $\sigma$-algebra $\Rightarrow \sigma$-ring $\Rightarrow$ ring but the converse is not true

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# Limit of Sequence of Sets:

**Definition:** Let $\{A_n\}_n^{\infty}$ be a sequence of sets in a universal set $X$.

**Limit inferior (liminf):** The limit inferiors of a sequence of sets is denoted by $\lim_{n\to\infty} \inf A_n$ and defined by

$$\lim_{n\to\infty} \inf A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$$

**Limit Supremum (lim.sup):** The limit sup of a sequence of sets is denoted by $\lim_{n\to\infty} \sup A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$.

if $\lim_{n\to\infty} \inf A_n = \lim_{n\to\infty} \sup A_n$ we say that limit of sequence of sets is exists and equal

$$\lim_{n\to\infty} A_n = \lim_{n\to\infty} \inf A_n = \lim_{n\to\infty} \sup A_n.$$

**Ex:** Find the limit of increasing sequence

$$A_n = \{1, 2, 3 \ldots n\} \subset N$$

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